Single Layer Perceptron and Adaline Classification Report:-

1. Perceptron Results

1.1 Summary table

No.	Feature 1	Feature 2	Class 1	Class 2	Bias	Acc. (L=0.01)	Acc. (L=0.1)	Acc. (L=0.001)
1	Culmen Length	Culmen Depth	Adelie	Chinstrap	With / without	95% / 95%	97.5% / 95%	100% / 100%
2	Culmen Length	Flipper Length	Adelie	Gentoo	With / without	100% / 100%	100% / 100%	100% / 97.5%
3	Culmen Depth	Origin Location	Chinstrap	Gentoo	With / without	100% / 100%	100% / 100%	100% / 100%
4	Flipper Length	Body Mass	Adelie	Chinstrap	With / without	70% / 65%	70% / 67.5%	72.5% / 65%
5	Origin Location	Body Mass	Adelie	Gentoo	With / without	80% / 90%	82.5% / 90%	85% / 85%

Each accuracy cell shows the result with bias / without bias. Learning rates: L = 0.01, 0.1, 0.001.

Perceptron results (test accuracy, 40 test samples, seed 42, 100 epochs max)

No.	Feature 1	Feature 2	Classes	Bias	Acc L=0.01	Acc L=0.1	Acc L=0.001
1	CulmenLength	CulmenDepth	Adelie vs Chinstrap	with	95.0%	97.5%	100.0%
1	CulmenLength	CulmenDepth	Adelie vs Chinstrap	without	95.0%	95.0%	100.0%
2	CulmenLength	FlipperLength	Adelie vs Gentoo	with	100.0%	100.0%	100.0%
2	CulmenLength	FlipperLength	Adelie vs Gentoo	without	100.0%	100.0%	97.5%
3	CulmenDepth	OriginLocation	Chinstrap vs Gentoo	with	100.0%	100.0%	100.0%
3	CulmenDepth	OriginLocation	Chinstrap vs Gentoo	without	100.0%	100.0%	100.0%
4	FlipperLength	BodyMass	Adelie vs Chinstrap	with	70.0%	70.0%	72.5%
4	FlipperLength	BodyMass	Adelie vs Chinstrap	without	65.0%	67.5%	65.0%
5	OriginLocation	BodyMass	Adelie vs Gentoo	with	80.0%	82.5%	85.0%
5	OriginLocation	BodyMass	Adelie vs Gentoo	without	90.0%	90.0%	85.0%

Figure 1: Perceptron results printed from the code (test accuracy for every combination, bias setting and learning rate).

1.2 Combination 1: Culmen Length vs Culmen Depth (Adelie vs Chinstrap)

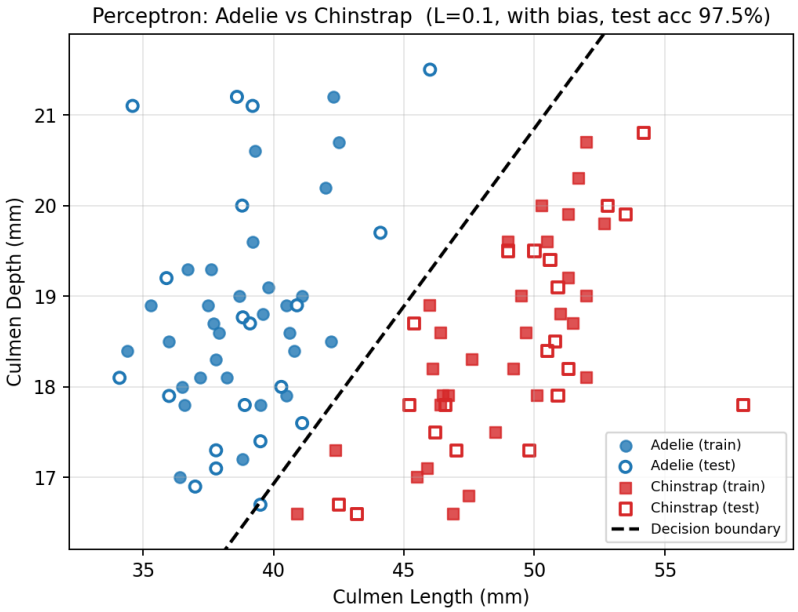


Figure 2: Perceptron decision boundary, Culmen Length vs Culmen Depth (Adelie vs Chinstrap). Filled markers = training samples, hollow = test samples. (L = 0.1, with bias).

Analysis: With bias the Perceptron reached 95%, 97.5% and 100% for L = 0.01, 0.1 and 0.001, and without bias 95%, 95% and 100% (at most two test errors). The two classes are therefore (almost) linearly separable in this feature space, and training converged in only 2 to 3 epochs at L = 0.1. Chinstrap penguins have much longer culmens than Adelie (about 48.9 mm against 38.8 mm on average), so Culmen Length does nearly all of the discriminating, while Culmen Depth is almost identical for the two species (18.5 mm against 18.8 mm) and only helps to tilt the line. The small differences between learning rates come from where the boundary ends up when training stops, not from a change in the data. This is a good performance example.

Feature discrimination: Culmen Length discriminates well between Adelie and Chinstrap (average 38.8 mm against 48.9 mm), while Culmen Depth does not (18.8 mm against 18.5 mm, almost identical). The separation therefore comes from Culmen Length, and the second feature only adds a small correction.

1.3 Combination 2: Culmen Length vs Flipper Length (Adelie vs Gentoo)

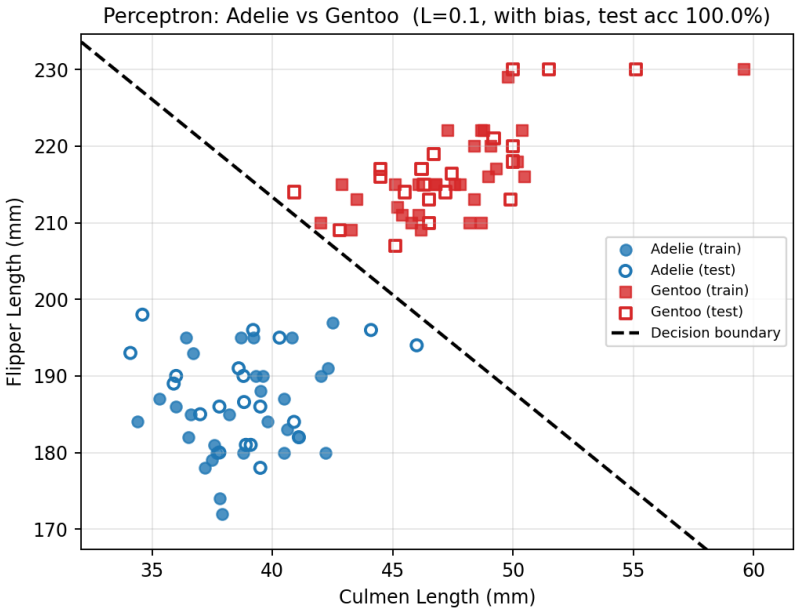


Figure 3: Perceptron decision boundary, Culmen Length vs Flipper Length (Adelie vs Gentoo). (L = 0.1, with bias).

Analysis: With bias this pair reached 100% at every learning rate, and without bias 100%, 100% and 97.5% for L = 0.01, 0.1 and 0.001. Gentoo penguins have clearly longer flippers than Adelie (every Gentoo has a flipper of at least 207 mm, every Adelie at most 198 mm) and also tend to have longer culmens, so the two clusters sit far apart with a visible gap. Training converged in 2 epochs. The single-sample difference without bias only reflects where the boundary lands inside that gap. This is a good performance example.

Feature discrimination: Both features discriminate well between Adelie and Gentoo. Flipper Length alone almost separates them (every Gentoo is at least 207 mm, every Adelie at most 198 mm), and Culmen Length adds extra separation, so the two classes form two distinct clusters.

1.4 Combination 3: Culmen Depth vs Origin Location (Chinstrap vs Gentoo)

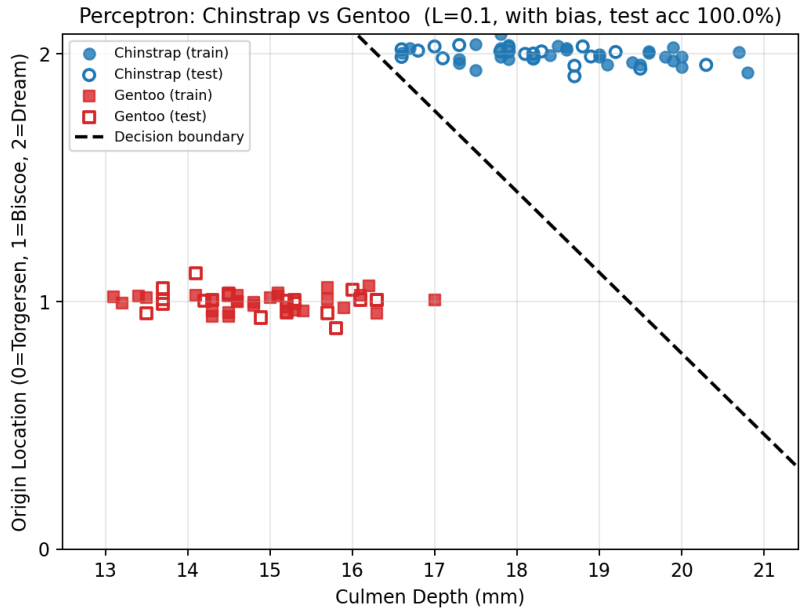


Figure 4: Perceptron decision boundary, Culmen Depth vs Origin Location (Chinstrap vs Gentoo). Points are jittered horizontally/vertically on the island axis for visibility. (L = 0.1, with bias).

Analysis: This was the most robust combination, with 100% accuracy for every learning rate, with and without bias. Gentoo penguins have shallower culmens than Chinstrap (about 14.9 mm against 18.5 mm), and in this dataset the two species live on different islands (all Chinstrap on Dream, all Gentoo on Biscoe), so Origin Location splits them perfectly. Note that part of this success comes from the island acting as a proxy for the species rather than from a biological measurement. The Perceptron converged after 2 epochs. This is a good performance example.

Feature discrimination: Both features discriminate between Chinstrap and Gentoo. Origin Location separates them perfectly in this dataset (Chinstrap on Dream, Gentoo on Biscoe), and Culmen Depth is also strongly different (14.9 mm against 18.5 mm), with only a small overlap.

1.5 Combination 4: Flipper Length vs Body Mass (Adelie vs Chinstrap)

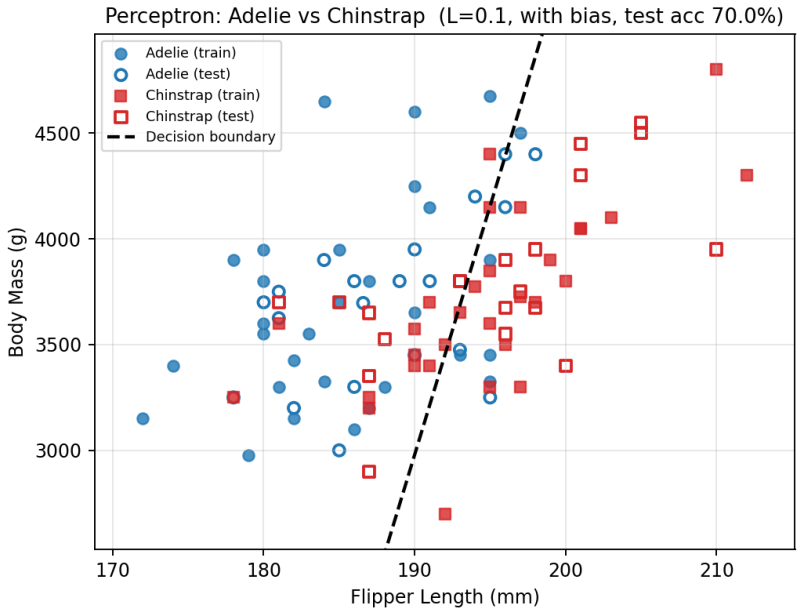


Figure 5: Perceptron decision boundary, Flipper Length vs Body Mass (Adelie vs Chinstrap). (L = 0.1, with bias).

Analysis: Accuracy ranged from 65% to 72.5%. With bias the results were 70%, 70% and 72.5% for L = 0.01, 0.1 and 0.001, and without bias 65%, 67.5% and 65%, so here the bias helped by a few points. Adelie and Chinstrap have almost the same body mass (about 3,700 g) and only slightly different flipper lengths, so the two clouds overlap heavily. When the classes are not linearly separable the Perceptron keeps updating the boundary and never settles (it used all 100 epochs in every run), so no learning rate can fix it. This is a bad performance example: these two features do not discriminate between these two classes.

Feature discrimination: Neither feature discriminates well between Adelie and Chinstrap: Body Mass is almost the same (about 3,696 g against 3,744 g) and Flipper Length differs only slightly (186.6 mm against 194.9 mm). The classes overlap heavily, so no line can separate them.

1.6 Combination 5: Origin Location vs Body Mass (Adelie vs Gentoo)

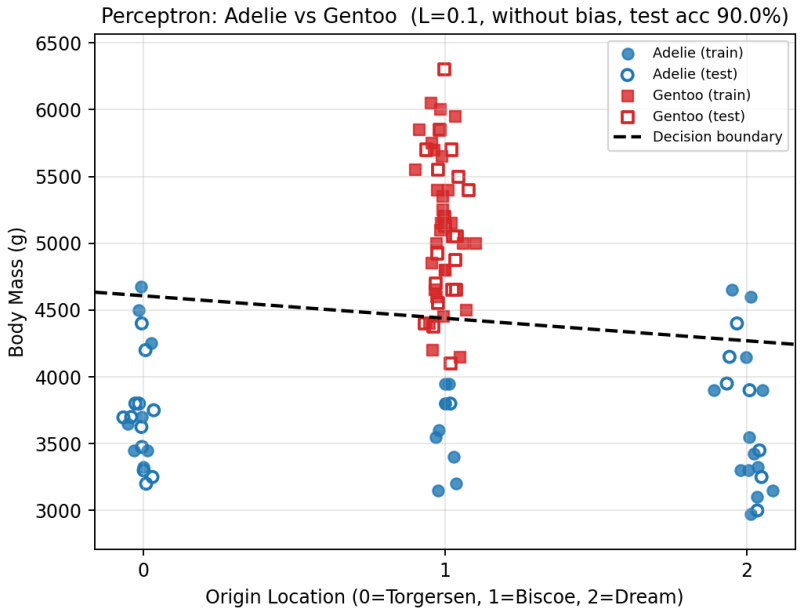


Figure 6: Perceptron decision boundary, Origin Location vs Body Mass (Adelie vs Gentoo). Points are jittered on the island axis for visibility. (L = 0.1, without bias).

Analysis: Accuracy ranged from 80% to 90%. Without bias the results were 90%, 90% and 85% for L = 0.01, 0.1 and 0.001, and with bias 80%, 82.5% and 85%, so here removing the bias helped. Gentoo penguins are much heavier (about 5,100 g against 3,700 g), which does most of the work, but Adelie penguins live on all three islands, including Biscoe where every Gentoo lives, so Origin Location cannot separate them cleanly. In addition, the island code (0 = Torgersen, 1 = Biscoe, 2 = Dream) is not an ordered measurement: Gentoo sit at the middle value, so no straight line on Origin Location alone can put Biscoe on one side and both Torgersen and Dream on the other. The boundary in the figure is almost horizontal, i.e. the decision is made almost only from Body Mass, and the heavier Adelie are misclassified. The Perceptron never converged (100 epochs). This is a medium performance example.

Feature discrimination: Body Mass discriminates fairly well between Adelie and Gentoo (about 3,696 g against 5,128 g, overlapping only between roughly 4,100 g and 4,675 g), but Origin Location discriminates only partly, because Adelie live on all three islands, including Biscoe where all Gentoo live.

2. Adaline Results

Adaline was tested on the same five combinations. Training with L = 1 made the cost explode (it overflowed to infinity in most settings, see Figure 10), so smaller values were required. Results are reported for L = 0.01, 0.1 and 0.001, and the plots below use L = 0.1 with bias.

2.1 Summary table

No.	Feature 1	Feature 2	Class 1	Class 2	Bias	Acc. (L=0.01)	Acc. (L=0.1)	Acc. (L=0.001)
1	Culmen Length	Culmen Depth	Adelie	Chinstrap	With / without	100% / 100%	100% / 100%	100% / 100%
2	Culmen Length	Flipper Length	Adelie	Gentoo	With / without	100% / 100%	100% / 100%	100% / 100%
3	Culmen Depth	Origin Location	Chinstrap	Gentoo	With / without	100% / 100%	100% / 100%	100% / 100%
4	Flipper Length	Body Mass	Adelie	Chinstrap	With / without	72.5% / 67.5%	70% / 67.5%	65% / 65%
5	Origin Location	Body Mass	Adelie	Gentoo	With / without	92.5% / 90%	87.5% / 90%	92.5% / 92.5%

Each accuracy cell shows the result with bias / without bias. Learning rates: L = 0.01, 0.1, 0.001.

Adaline results (test accuracy, 40 test samples, seed 42, 100 epochs max)

No.	Feature 1	Feature 2	Classes	Bias	Acc L=0.01	Acc L=0.1	Acc L=0.001	Acc L=1
1	CulmenLength	CulmenDepth	Adelie vs Chinstrap	with	100.0%	100.0%	100.0%	57.5%
1	CulmenLength	CulmenDepth	Adelie vs Chinstrap	without	100.0%	100.0%	100.0%	90.0%
2	CulmenLength	FlipperLength	Adelie vs Gentoo	with	100.0%	100.0%	100.0%	5.0%
2	CulmenLength	FlipperLength	Adelie vs Gentoo	without	100.0%	100.0%	100.0%	90.0%
3	CulmenDepth	OriginLocation	Chinstrap vs Gentoo	with	100.0%	100.0%	100.0%	50.0%
3	CulmenDepth	OriginLocation	Chinstrap vs Gentoo	without	100.0%	100.0%	100.0%	100.0%
4	FlipperLength	BodyMass	Adelie vs Chinstrap	with	72.5%	70.0%	65.0%	55.0%
4	FlipperLength	BodyMass	Adelie vs Chinstrap	without	67.5%	67.5%	65.0%	35.0%
5	OriginLocation	BodyMass	Adelie vs Gentoo	with	92.5%	87.5%	92.5%	87.5%
5	OriginLocation	BodyMass	Adelie vs Gentoo	without	90.0%	90.0%	92.5%	60.0%

Figure 7: Adaline results printed from the code (test accuracy for every combination, bias setting and learning rate, including L = 1).

2.2 Combination 1: Culmen Length vs Culmen Depth (Adelie vs Chinstrap)

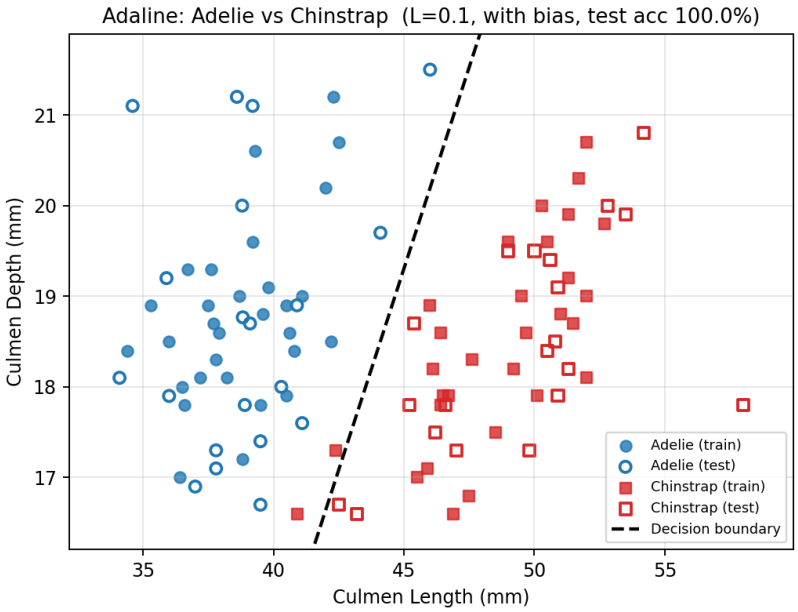


Figure 8: Adaline decision boundary, Culmen Length vs Culmen Depth (Adelie vs Chinstrap) (L = 0.1, with bias).

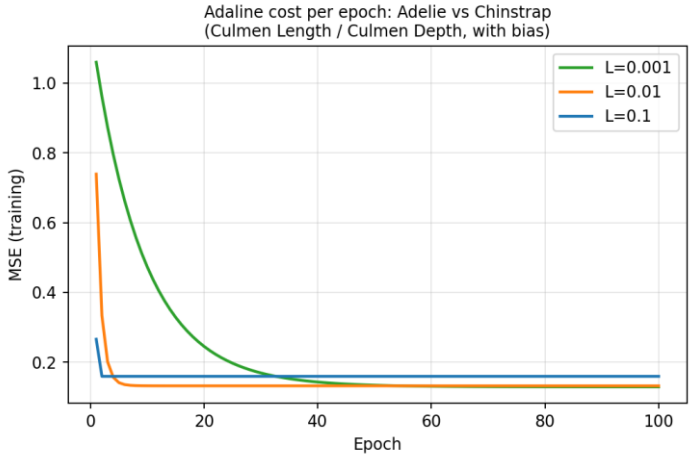


Figure 9: Adaline training MSE per epoch for the learning rates 0.001, 0.01 and 0.1 (with bias).

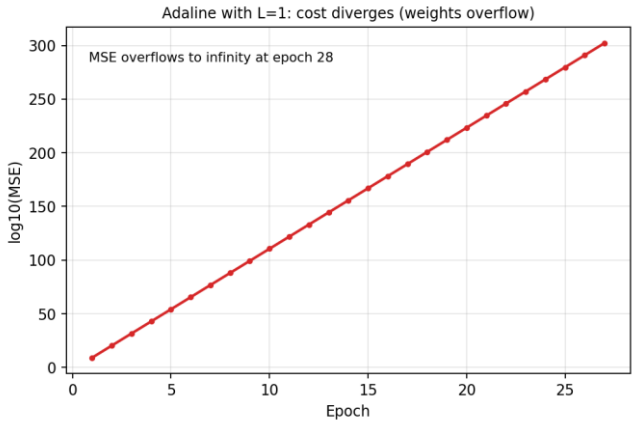


Figure 10: Adaline with L = 1 on the same combination: the cost diverges and overflows.

Analysis: Adaline reached 100% test accuracy at all three learning rates, with and without bias. The classes are well separated by a line, and the cost curve drops quickly in the first few epochs and then flattens. A larger L converges faster but settles at a slightly higher MSE (about 0.16 at L = 0.1 against 0.13 at L = 0.001), because each per-sample update is larger. The plateau stays above zero because Adaline's output is linear while the targets are +1/-1; the accuracy only uses the sign of the output. With L = 1 the updates were far too large: the cost grew by roughly ten orders of magnitude per epoch and overflowed after 28 epochs (with bias). This is a good performance example.

Feature discrimination: Culmen Length discriminates well between Adelie and Chinstrap (average 38.8 mm against 48.9 mm), while Culmen Depth does not (18.8 mm against 18.5 mm, almost identical). The separation therefore comes from Culmen Length, and the second feature only adds a small correction.

2.3 Combination 2: Culmen Length vs Flipper Length (Adelie vs Gentoo)

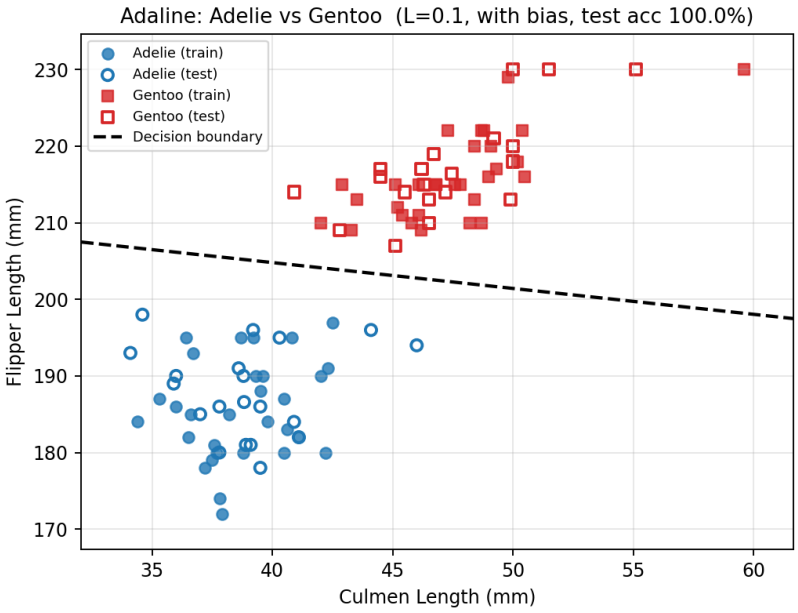


Figure 11: Adaline decision boundary, Culmen Length vs Flipper Length (Adelie vs Gentoo) (L = 0.1, with bias).

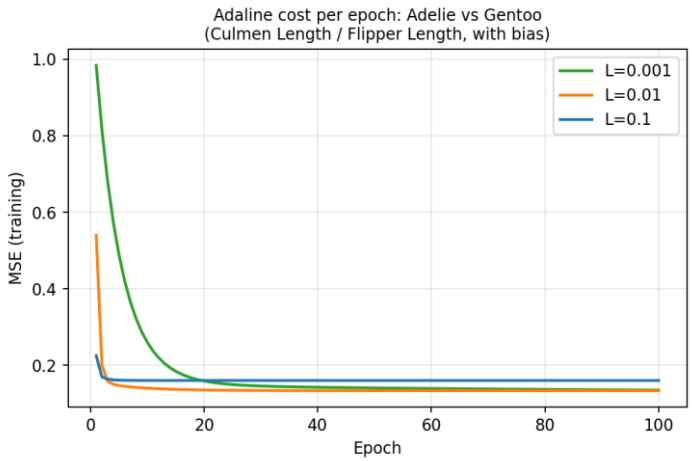


Figure 12: Adaline training MSE per epoch for the learning rates 0.001, 0.01 and 0.1 (with bias).

Analysis: 100% accuracy at all three learning rates, with and without bias. Gentoo is clearly larger than Adelie in flipper length and also longer in culmen length, so the clusters are far apart and a single line separates them. The cost flattens around 0.13 to 0.16, which is again the floor for a linear output against +1/-1 targets. L = 1 diverged here as well (the cost overflowed after 25 epochs with bias), so its accuracy is not meaningful. This is a good performance example.

Feature discrimination: Both features discriminate well between Adelie and Gentoo. Flipper Length alone almost separates them (every Gentoo is at least 207 mm, every Adelie at most 198 mm), and Culmen Length adds extra separation, so the two classes form two distinct clusters.

2.4 Combination 3: Culmen Depth vs Origin Location (Chinstrap vs Gentoo)

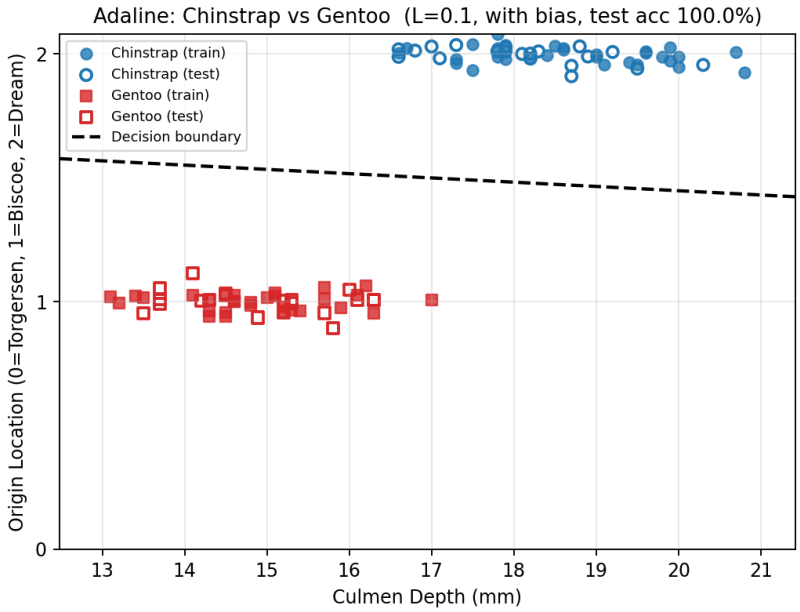


Figure 13: Adaline decision boundary, Culmen Depth vs Origin Location (Chinstrap vs Gentoo) (L = 0.1, with bias).

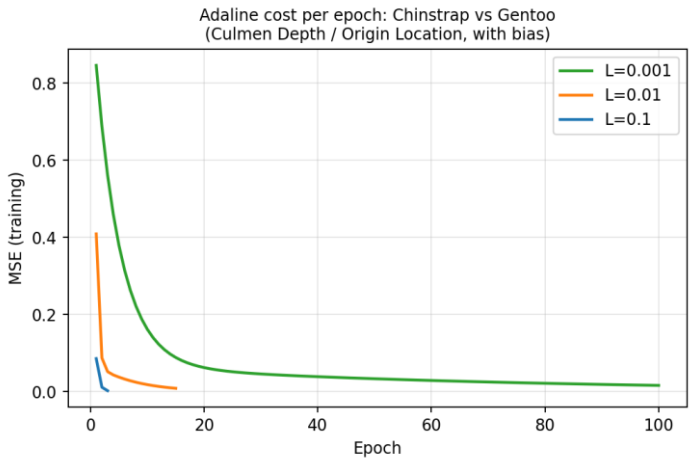


Figure 14: Adaline training MSE per epoch for the learning rates 0.001, 0.01 and 0.1 (with bias).

Analysis: 100% accuracy at every learning rate, with and without bias. As with the Perceptron, the different island and the different culmen depth of the two species make them very easy to separate. The cost falls fastest here: it dropped below the 0.01 stopping threshold after only 3 epochs at L = 0.1 and 15 epochs at L = 0.01, and reached 0.016 after 100 epochs at L = 0.001. This is a good performance example.

Feature discrimination: Both features discriminate between Chinstrap and Gentoo. Origin Location separates them perfectly in this dataset (Chinstrap on Dream, Gentoo on Biscoe), and Culmen Depth is also strongly different (14.9 mm against 18.5 mm), with only a small overlap.

2.5 Combination 4: Flipper Length vs Body Mass (Adelie vs Chinstrap)

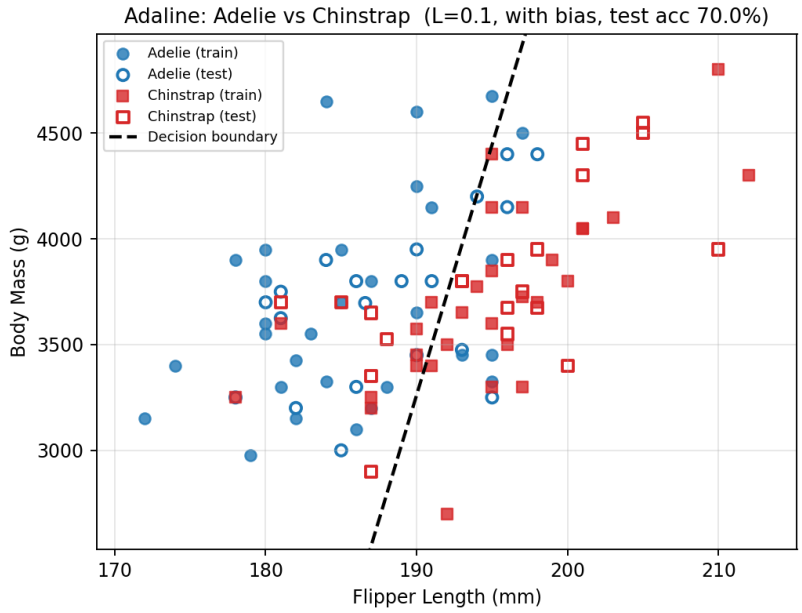


Figure 15: Adaline decision boundary, Flipper Length vs Body Mass (Adelie vs Chinstrap) (L = 0.1, with bias).

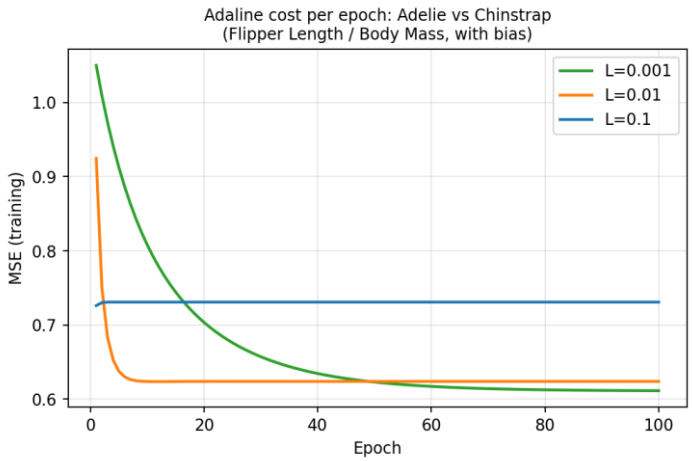


Figure 16: Adaline training MSE per epoch for the learning rates 0.001, 0.01 and 0.1 (with bias).

Analysis: Accuracy was 72.5%, 70% and 65% with bias (L = 0.01, 0.1, 0.001) and 67.5%, 67.5% and 65% without bias. The two classes overlap heavily on these features, so even the best line misclassifies many test points. The cost curve flattens at a high value (about 0.61 to 0.73), meaning the error cannot be reduced further by training longer. The result is the same as the Perceptron's best (72.5%). This is a bad performance example.

Feature discrimination: Neither feature discriminates well between Adelie and Chinstrap: Body Mass is almost the same (about 3,696 g against 3,744 g) and Flipper Length differs only slightly (186.6 mm against 194.9 mm). The classes overlap heavily, so no line can separate them.

2.6 Combination 5: Origin Location vs Body Mass (Adelie vs Gentoo)

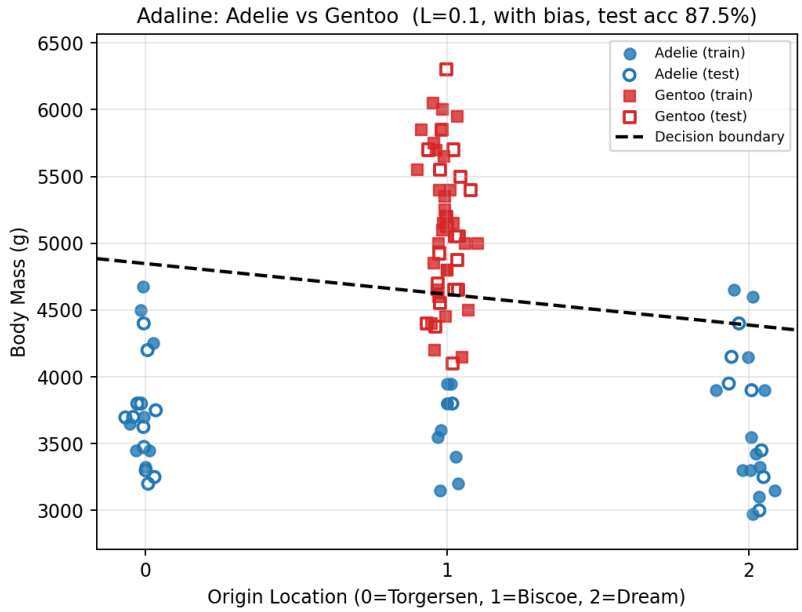


Figure 17: Adaline decision boundary, Origin Location vs Body Mass (Adelie vs Gentoo) (L = 0.1, with bias).

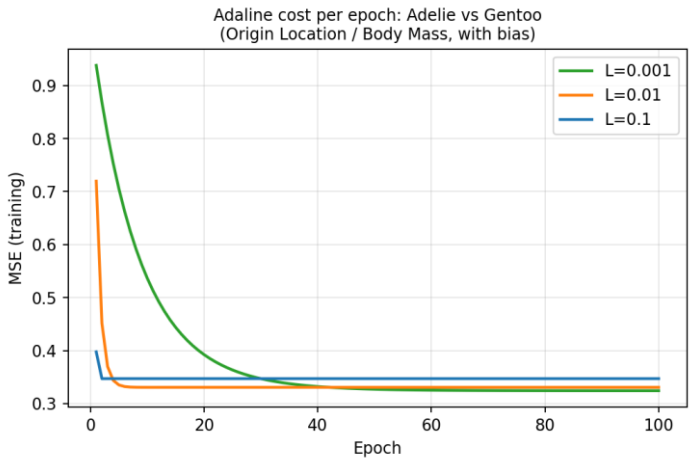
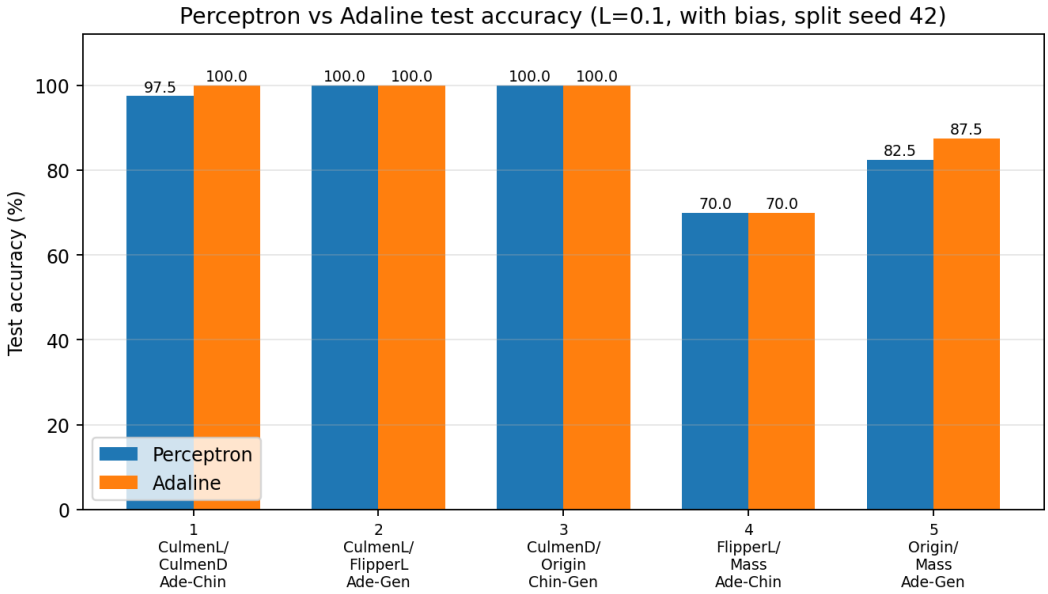


Figure 18: Adaline training MSE per epoch for the learning rates 0.001, 0.01 and 0.1 (with bias).

Analysis: Accuracy was 92.5%, 87.5% and 92.5% with bias, and 90%, 90% and 92.5% without bias. Body Mass separates most Gentoo from Adelie, but the overlap caused by Adelie living on the same island as Gentoo, together with the unordered island code, prevents a perfect result. The cost flattens at about 0.32 to 0.35. This is a medium performance example.

Feature discrimination: Body Mass discriminates fairly well between Adelie and Gentoo (about 3,696 g against 5,128 g, overlapping only between roughly 4,100 g and 4,675 g), but Origin Location discriminates only partly, because Adelie live on all three islands, including Biscoe where all Gentoo live.

3. Comparison Between the Algorithms



4. Conclusion

The highest accuracy, 100%, was achieved by three feature pairs by both the Perceptron and Adaline :-

No.	Feature pair	Classes	Best Perceptron	Best Adaline	Performance
1	Culmen Length + Culmen Depth	Adelie vs Chinstrap	100%	100%	Good
2	Culmen Length + Flipper Length	Adelie vs Gentoo	100%	100%	Good
3	Culmen Depth + Origin Location	Chinstrap vs Gentoo	100%	100%	Good (most robust)

The most robust pair was Culmen Depth with Origin Location, which gave 100% in every Perceptron and Adaline setting with L up to 0.1 (and a mean of 99.9% for the Perceptron and 100.0% for Adaline over 30 random splits), partly because the island acts as a proxy for the species. Culmen Length appeared in two of the three best pairs, so it is the most useful single feature across species. The worst pair was Flipper Length with Body Mass for Adelie vs Chinstrap (65% to 72.5%), because those two species have almost the same body mass and similar flippers, so these features do not discriminate between them.